

Second-order correlation for the classical and quantum light sources using Timepix Camera

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Abstract: We implemented an experimental aspect of the second-order intensity-intensity correlation function for a coherent light source, employing the high-resolution Timepix (Tpx3Cam) camera to precisely measure photon arrival times and spatial distribution.

1. Introduction

Accurate and timely determination of photon arrival times plays a critical role across a broad spectrum of disciplines, including physics, chemistry, and the life sciences. The Hanbury Brown and Twiss (HBT) interferometer [1, 2], despite its initial controversies, has become a foundational tool in photon correlation measurements. Simultaneous detection of photons provides vital information about the nature of the light source, revealing properties such as coherence, statistical behavior, and quantum characteristics [3]. In particular, the coincident arrival of photons at spatially separated detectors can offer insight into non-local correlations and quantum entanglement. In nuclear physics, coincident particle detection enables the reconstruction of decay pathways and enhances the understanding of nuclear processes. However, recording the individual arrival times of quantum particles over extended durations—across single or multiple detection channels—yields more granular data and facilitates a deeper exploration of the underlying physical mechanisms.

In this study, we employed the Tpx3Cam camera (<https://www.amscins.com/tpx3cam/>) to investigate second-order correlations for the coherent light source and benchmarked its performance against a conventional time-to-digital converter (Swabian Timetagger 20). The Tpx3Cam has gained prominence in quantum optics due to its exceptional temporal (1.6 ns) and spatial ($55 \times 55 \mu\text{m}^2$) resolution, making it highly effective for characterizing entangled photon sources. Its capabilities allow for precise measurement of photon timing, correlations, and spatial distributions, thereby contributing significantly to the analysis and optimization of quantum photonic systems [4, 5]. Unlike traditional imaging technologies such as EMCCD, CMOS, or ICCD cameras, the Tpx3Cam features a data-driven readout architecture that supports a high throughput of up to 10 million photons per second, positioning it as a transformative tool for advanced quantum optics experiments.

2. Second-order correlation using Tpx3Cam

The Tpx3Cam camera, when coupled with an image intensifier (intensified Tpx3Cam), provides single-photon sensitivity and enables precise time-stamping of incident photons with a temporal resolution of 1.6 ns. This system comprises a light-sensitive silicon sensor that is bump-bonded to the Timepix3 readout ASIC, a time-stamping chip. The sensor-chip assembly features a matrix of 256×256 pixels, each measuring $55 \times 55 \mu\text{m}$. Each pixel in the Timepix3 chip functions independently and includes processing electronics that record the time of arrival (TOA) of signals exceeding a preset threshold. These are timestamped data, stored as time codes in each individual pixel memory. In addition to TOA, the time-over-threshold (TOT) is recorded, which provides an estimate of the photon flux incident on that pixel. The dead time for each pixel, caused by digitization and TOT measurement, is on the order of $1 \mu\text{s}$. The Timepix3 operates with a data-driven, event-based readout scheme: only those pixels registering signals above the threshold are read out. This allows the camera to function continuously without the need for a global trigger, as each pixel asynchronously transfers data within microseconds after activation. The camera supports a maximum throughput of approximately 80 megapixels per second. Additionally, the camera interface can accept and time-stamp an external trigger pulse independently of the Timepix3 data stream. We recorded the TOA data from the experiment at two different locations of the sensor of Tpx3Cam, as shown in Fig. 1.

We carried out time ordering, and clustering of the data, and isolated all data from the rest of the pixels and stamped (labeled) it into channel 1 and channel 2.

All data is converted into real-time and arranged in ascending order of all the channels. We made the timing histogram from that data by subtracting the two-channel real-time columns, defining the first channel as a START pulse and the second channel as a STOP pulse, and plotting it as a time histogram or correlation histogram. Once we have the timing histogram or correlation histogram, then it will be normalized using the given formula for getting the second-order correlation $g^2(\tau)$ function, such as

$$g^2(\tau) = \frac{\Delta t}{\text{binwidth}(\tau) \cdot N_1 \cdot N_2} \cdot \text{Histogram}(\tau). \quad (1)$$

Where Δt is the total time duration of taking data, binwidth is the coincidence window, and N_1 and N_2 are the total count rates in channel 1 and channel 2, respectively. It is well known that the value of $g^2(\tau)$ at $\tau = 0$ comes out to be 2, 1, and $1-1/n$ for thermal, coherent, and nonclassical states (with n photons).

3. Results and Discussion

We employed the He-Ne laser for the detection of the second-order intensity-intensity correlation $g^2(\tau)$ using the Tpx3Cam, and its result can be seen in Fig. 2. We took the data for 20 sec total time with the exposure time of 1 sec (999 ms + 1 ms dead time) and 1 frame rate per second. As expected for a coherent light source, the correlation function maintains a unit value across all delays and shows excellent agreement with measurements obtained using the conventional Swabian Time Tagger 20.

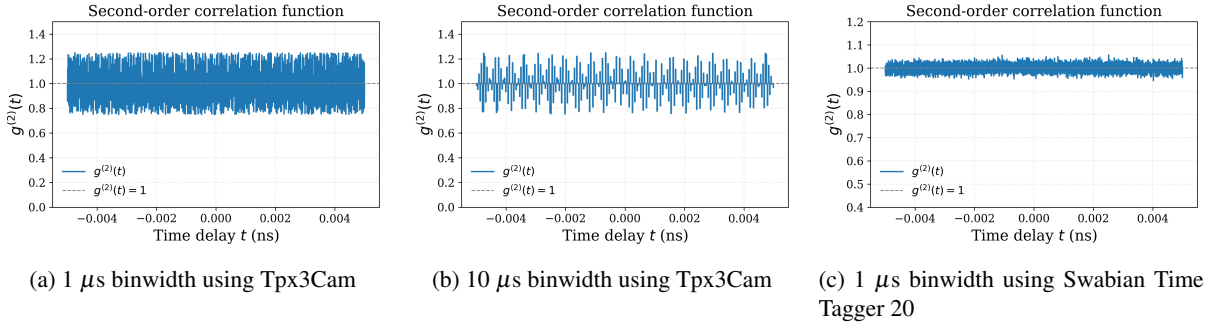


Fig. 2: Plot of the intensity-intensity correlation for the coherent light obtained from the Tpx3Cam and Swabian TimeTagger 20. All plots have the $g^2(\tau)$ value of one as expected for the coherent light.

4. Conclusion

We employed the Tpx3Cam for the investigation of second-order intensity-intensity correlation $g^2(\tau)$ for the coherent light source. We got the expected unity values for all the delays and compared them with the Swabian Time Tagger 20. In the future, we will investigate it for all the light sources, such as thermal and quantum light sources.

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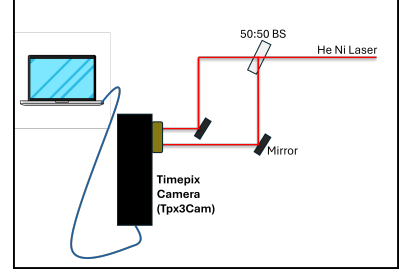


Fig. 1: Experimental layout for the investigation of $g^2(\tau)$ using Tpx3Cam.